

Weekly Site Report

BLACK ROCK WEEKLY REPORT

Weekly PSD Report
EPIROC SA

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Weekly Service Report – Black Rock Mine Operations

Report type:	Weekly	Submitted to:	Sello Taku and Sipho
Week Date:	28 November 2025	End Date	4 December 2025
Reported By:	Philip Moller		
BEV:			
Service Tech:	Ivan Phiri (morning shift)	Service Tech:	Lukas Serumola (Night shift)
Service Tech:	04 Dec and Friday 05 Dec for a funeral.	Service Tech Assistant:	Patrick Yoko (Batteries)
Specialist:	Frikkie van der Walt (Batteries)		
Field Service:			
Service Tech:	Lucas Slaffa	Service Tech:	Mathews Mosemane
CAS Project:			
Team Leader:	Wisani Shilenge	Service Tech:	Phineas Sibanda
Service Tech:	Julius Manamela	Service Tech:	Joseph Mampa
Certiq Project:			
Team Leader:	Segunda Fernando	Service Tech:	
Mobilaris:			
Team Leader:	Rupert Grobler		

Report Summary

General Summary

The purpose of this report is to give an overview of all the Epiroc activities at Blackrock Mine Operations (BRMO).

BEV - Critical Breakdown Summary

- Continuing with audit on battery disconnects and compatibility. Photo report on connector conditions carried out and was shared with Sello and Piet.
- We had 5 batteries on breakdown start of week. 2 loader batteries and 3 truck batteries.
VPY00011 – PSB (charging PC board) changed, need to upload parameters.

VPY00049 – Level 3 fault on a sub pack. Scania reset sub pack

VPX00023 - Level 3 fault on a sub pack. Scania reset sub pack

VPX00026 - Level 3 fault on a sub pack. Waiting on Scania

VPX00031 - Level 3 fault on a sub pack. Waiting on Scania

There is a bug in the software that is locking out the battery subpacks due to an over current fault. Scania assisting in unlocking the sub packs and going through data to make sure there are no other issues before resetting packs.

- Battery bay crane on breakdown since 02/12/2025 (Tuesday). Waiting on brake parts. Operating with crane but is very slow. Started using crane 03/12/2025 (Wednesday) at 14:00.
- DT149 and DT146 long downs due to Strata breakdowns.

Outstanding audits:

DT162

FL113

CAS L9 Implementation - Critical Issues Summary

- Done, in sign off stage.

Certiq & Mobilaris - Critical Issues Summary

Passport 360 Compliance

- BEV Technicians: **96%**
- CASL9 (Gloria): **100%**
- CASL9 (Nch2): **98%**
- CAS L9 (Nch3): **98%**
- Certiq & Mobilaris N2: **92%**
- Certiq & Mobilaris Gloria: **83% -**

BEV Report

Weekly machine utilization summary:

Friday 28 November 2025:

1. FL 98 -
2. FL 99 -
3. FL 107 -
4. FL 108 -
5. FL 112 -
6. FL 113 -
7. DT 146 – Morning shift: 3rd party supplier fault – Strata. (1H34min)
8. DT 147 -
9. DT 149 - Morning shift: 3rd party supplier fault - Strata (3H37min)
Night shift: 3rd party supplier fault - Strata (9H18min)
10. DT 150 -
11. DT 162 -
12. DT 163 - Morning shift: 3rd party supplier fault - Strata (2H26min)
13. DT 171 –

Saturday 29 November 2025:

1. FL 98 -
2. FL 99 -
3. FL 107 -
4. FL 108 -
5. FL 112 - Morning shift: 3rd party supplier fault - Strata (43min)
6. FL 113 -
7. DT 146 -
8. DT 147 -
9. DT 149 -
10. DT 150 -
11. DT 162 -
12. DT 163 -
13. DT 171 -

Sunday 30 November 2025:

1. FL 98 -
2. FL 99 -
3. FL 107 -
4. FL 108 -
5. FL 112 - Morning shift: 3rd party supplier fault - Strata (23H02min)
6. FL 113 -
7. DT 146 -
8. DT 147 -
9. DT 149 -
10. DT 150 -
11. DT 162 -
12. DT 163 - Morning shift: 3rd party supplier fault - Strata (19H19min)
13. DT 171 -

Monday 01 December 2025:

1. FL 98 -
2. FL 99 - Afternoon shift: Mechanical breakdown – Brake system – No feedback. (55min)
3. FL 107 -
4. FL 108 -
5. FL 112 -
6. FL 113 - Afternoon shift: Tyre Bay breakdown – Changed tyre. (4H34min)
7. DT 146 - Morning shift: 3rd party supplier fault - Strata (36min)
8. DT 147 -
9. DT 149 - Morning shift: 3rd party supplier fault - Strata (1D3H29min)
10. DT 150 -
11. DT 162 -
12. DT 163 -
13. DT 171 -

Tuesday 02 December 2025:

1. FL 98 - Night shift: Electrical breakdown – TCU error. Reset tripped fuse. (1H02min)
2. FL 99 -
3. FL 107 -
4. FL 108 -
5. FL 112 -
6. FL 113 -
7. DT 146 -
8. DT 147 -
9. DT 149 - Night shift: 3rd party supplier fault - Strata (2H30min)
10. DT 150 -
11. DT 162 -
12. DT 163 -
13. DT 171 -

Wednesday 03 December 2025:

1. FL 98 - Morning shift: Mechanical breakdown – Auto phasing fault. Did ID runs. (3H41min)
2. FL 99 -
3. FL 107 - Morning shift: 3rd party supplier fault – Strata (2H47min)
4. FL 108 -
5. FL 112 - Afternoon shift: Tyre-bay breakdown – Replaced tyre (1D13H57min)
6. FL 113 -
7. DT 146 - Night shift: 3rd party supplier fault - Strata (8H26min)
8. DT 147 -
9. DT 149 - Afternoon shift: 3rd party supplier fault – Strata (3H41min)
Morning shift: 3rd party supplier fault – Strata (4H40min)
10. DT 150 - Night shift: 3rd party supplier fault – Strata (5H29min)
11. DT 162 -
12. DT 163 -
13. DT 171 -

Thursday 04 December 2025:

14. FL 98 -
15. FL 99 -
16. FL 107 -
17. FL 108 -
18. FL 112 -
19. FL 113 -
20. DT 146 -
21. DT 147 -
22. DT 149 - Afternoon shift: 3rd party supplier fault - Ongoing fault (1D13H57min)
23. DT 150 -
24. DT 162 -
25. DT 163 -
26. DT 171 -

Re-occurring issues:

1. Strata

Improvement and actions:

1. DCDC campaign kicked off to reduce charger interruptions. (Ongoing)
4 batteries remaining to be converted this week.
2. Post 3,6 and 7 CCS connector cables need to be replaced. (Cause of High pin temp)
See below status of parts.

OPEN ORDER REPORT FOR BLACK ROCK: 12 NOVEMBER 2025

Order Date	Order Number	Item number	Description	Ordered	Shipped	Qty open	Status	ETA
2025/11/13	25001480/07/MW	9464900654	GROMMET	3	0	3	Open	2025/12/05
2025/11/13	25001480/07/MW	3180004087	SET,SPRING	3	0	3	Open	2025/12/05
2025/11/13	25001480/07/MW	3180004088	HOLDER	3	0	3	Open	2025/12/05
2025/11/13	25001480/07/MW	3180004092	KIT,SERVICE	3	0	3	Open	2025/12/09

3. **Auxiliary motor spline grease** TSNB not done. We need to schedule the machines to complete this campaign. Grease already on site. This have been raised multiple times; I need BR to schedule the machines.
4. Connector audits on Machines almost done. 2 machines to be audited. (FL113 and DT162) We will have to replace the connectors.
5. Regeneration knob set to 100% to be started this week.

SERVICES:

Nchwaning 3

Maintenance Schedule Dec 2025				
Monday	Tuesday	Wednesday	Thursday	Friday
8-Dec-25	9-Dec-25	10-Dec-25	11-Dec-25	12-Dec-25
DT0109	DT0147	DT0148	DT0149	
FL0099	FL0090		DZ0011	
HD0049	HD0050	HD0051	HD0052	HD0055

Battery and Charger Status Report

General summary

ST14 – B4 battery packs

- We currently have 10 x ST14 – B4 battery packs and 6 x ST14 machines operational underground. This is equal to the committed ratio of 1.6 batteries per machine.

MT42 – B5 battery packs

- We currently have 12 x MT42 – B5 battery packs operational underground. This is above the committed ratio of 1.6 batteries per machine.

160 kW Chargers

- Post 3,6 and 7 CCS connector cables need to be replaced. New 300A cables was order and received. Need to order upgraded spring set. Piet did order upgrade kit. Waiting parts.

General

- There are more noted charger stops; this is being investigated with initial indications being worn out charger and CCS battery inlet connectors. DCDC campaign initiated. 5 batteries remaining.

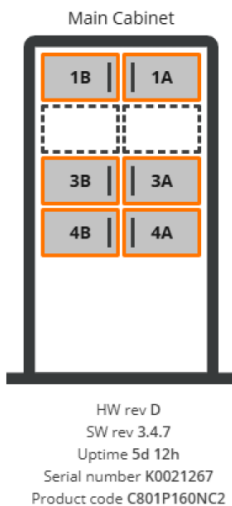
Table 1: Battery pack availability for week 2025

Battery Type	Battery ID	Status	Comment
B4 - ST14	VPY-00011	Breakdown	Changed PSB – Need to load software on PSB. A T-link cable is needed. Asked to be sent to the site.
B4 - ST14	VPY-00031	Working	
B4 - ST14	VPY-00051	Working	

B4 - ST14	VPY-00048	Working	
B4 - ST14	VPY-00049	Working	Was on breakdown – Scania reset a level 3 fault
B4 - ST14	VPY-00088	Working	
B4 - ST14	VPY-00086	Working	
B4 - ST14	VPY-00076	Working	
B4 - ST14	VPY-00083	Working	
B4 - ST14	VPY-00041	Working	
B5 - MT42	VPX-00016	Working	
B5 - MT42	VPX-00015	Working	
B5 - MT42	VPX-00023	Working	Was on breakdown – Scania reset a level 3 fault.
B5 - MT42	VPX-00017	Working	
B5 - MT42	VPX-00010	Working	
B5 - MT42	VPX-00050	Working	
B5 - MT42	VPX-00031	Breakdown	VPC fault (4) - waiting feedback from Scania
B5 - MT42	VPX-00036	Working	
B5 - MT42	VPX-00028	Working	
B5 - MT42	VPX-00026	Working	Need to change subpack
B5 - MT42	VPX-00024	Working	
B5 - MT42	VPX-00048	Working	
B5 - MT42	VPX-00044	Surface	On machine: 8997901500

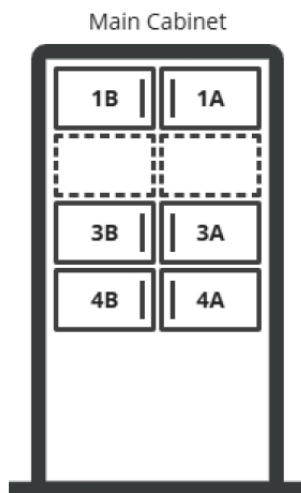
Battery charger summary:

Charger (BRM 1)



- Charger missing 2X power modules
- Charging power reduced
- Modules need to be replaced
- Charger currently running with no faults
- When Charger modules are highlighted orange, Means charger is currently charging
- Charger currently on new SW V3.4.7

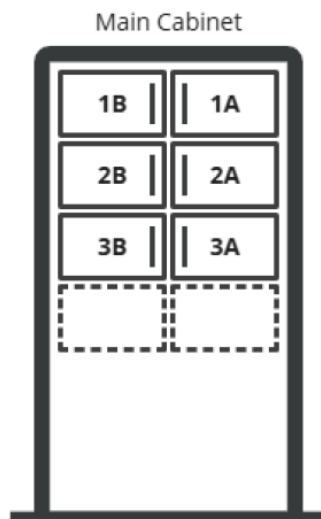
Charger (BRM 2)



HW rev D
SW rev 3.4.7
Uptime 17d 0h
Serial number K0021363
Product code C801P160NC2

- Charger missing 2X power modules
- Charging power reduced
- Modules need to be replaced
- Charger currently running with no faults
- Charger currently on new SW V3.4.7

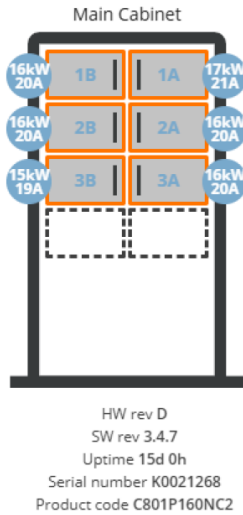
Charger (BRM 3)



HW rev D
SW rev 3.4.7
Uptime 9d 16h
Serial number K0021364
Product code C801P160NC2

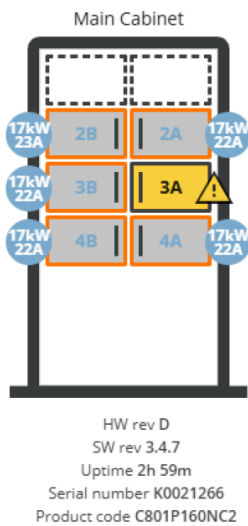
- Charger missing 2X power modules
- Charging power reduced
- Modules need to be replaced
- Charger currently running with no faults
- Charger currently on new SW V3.4.7

Charger (BRM 4)



- Charger missing 2X power modules
- Charging power reduced
- Modules need to be replaced
- Charger currently running with no faults
- When Charger modules are highlighted orange, Means charger is currently charging
- Charger currently on new SW V3.4.7

Charger (BRM 5)



ERRORS

POWER MODULE FAULT
MC 3A

- FAN_CURRENT
MC 3A
- RT_CONNECTOR_PMC_ERROR
MC 3A
- SSC_PMC_FAN_CURRENT
MC 3A

POWER MODULE FAULT
17.11.2025 13:32 (UTC+2)

PMC 3A on Main cabinet: SSC_PMC_FAN_CURRENT
Remote investigation required

POTENTIAL IMPACT
No information

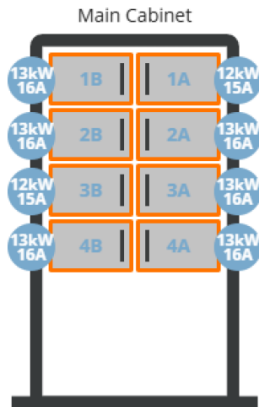
STEPS TO FIX
No information

DETAILS

Error code	CE_PMC_ERROR	Copy
Vendor	Kempower	
Occurred	17.11.2025 13:32 (UTC+2)	
Cause	No information	
Documentation	No information	

- Charger missing 2X power modules
- Charging power reduced
- Charger currently with a defective fan on the power module highlighted yellow, Charger fan needs to be replaced.
- Charger on new SW V3.4.7

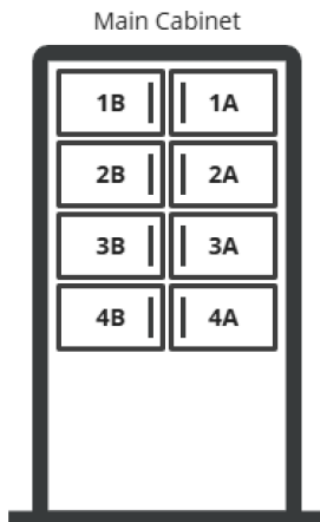
Charger (BRM 6)



- Charger currently running with no faults
- When Charger modules are highlighted orange, Means charger is currently charging
- Charger currently on new SW V3.4.7

HW rev D
SW rev 3.4.7
Uptime 9d 18h
Serial number K0021365
Product code C801P160NC2

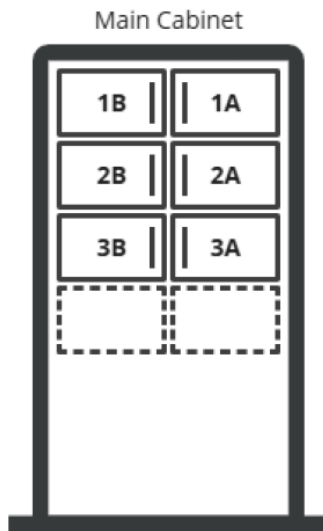
Charger (BRM 7)



- Charger currently running with no faults
- Charger currently on new SW V3.4.7

HW rev K
SW rev 3.4.7
Uptime 45m
Serial number K0028147
Product code C801P160ND4LC2

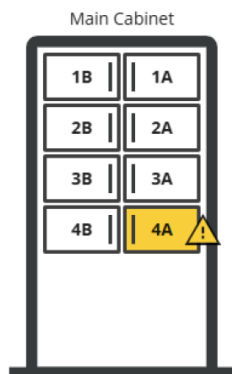
Charger (BRM 8)



HW rev K
SW rev 3.4.7
Uptime 12d 7h
Serial number K0028146
Product code C801P160ND4LC2

- Charger missing 2X power modules
- Charging power reduced
- Modules need to be replaced
- Charger currently running with no faults
- Charger currently on new SW V3.4.7

Charger (BRM 9)



HW rev L
SW rev 3.4.7
Uptime 15d 11h
Serial number K0032320
Product code C801MVP160ND4LC2

ERRORS

▲ POWER MODULE FAULT
MC 4A

▲ RT_PMC_CONTINUOUS_FAILURES
MC 4A

RT_PMC_CONTINUOUS_FAILURES

28.11.2025 12:09 (UTC+2)

Power Module Channel (PMC) has continuous errors.
▲ Remote investigation required

POTENTIAL IMPACT

Charging power unit's total power output reduced by 25 kW.

STEPS TO FIX

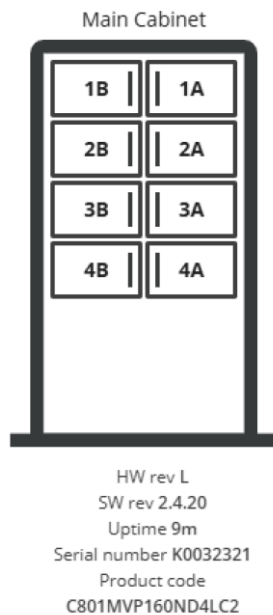
Do remote fault analysis with Service Partner or Kempower Technical Support.

DETAILS

Error code	RT_PMC_CONTINUOUS_FAILURES	Copy
Vendor	Kempower	
Occurred	28.11.2025 12:09 (UTC+2)	
Cause	This error appears after a runtime error has affected the Power Module channel (PMC) for more than 5 times by default.	
Documentation	No information	

- Charger with power module failure or module switched off
- Charger output reduced by 25KW
- Power module needs to be replaced if defective
- Charger on new SW version V3.4.7

Charger (BRM 10)



- Charger not connected to the Grid and offline
- Charger on old software and needs to be upgraded to SW V3.4.7

FST Report Summary

Field Service on site:

Mathews Mosemane – Assisting at BEV workshop.

Pontsho – Commissioning of HD0070 (New S2)

Julius Manamela – Fault finding on FL0092.

CAS Report Summary

Report Type:	Weekly	Project Leader:	Hennie van Niekerk
Start Date:	15-Nov-2025	Report By:	Bradley Nast
End Date:	21-Nov-2025	Contract No:	BPE145D

Critical Issues

Progress Update

All machines on Cas lvl 9.

Project in sign off phase.